

REAL-TIME OPERATING SYSTEM IN TRAFFIC SIGNAL CONTROL

Course Name: Operating Systems

Course Code: CSA0407

Assessment Tool: 3 – Technical Presentation

Presented by:

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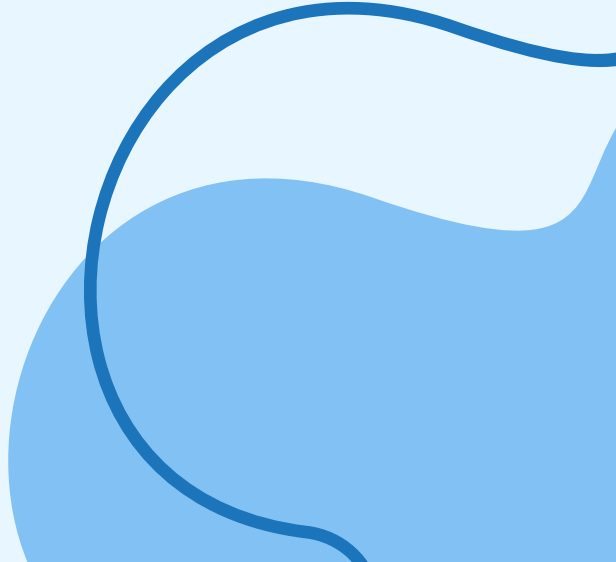


INTRODUCTION

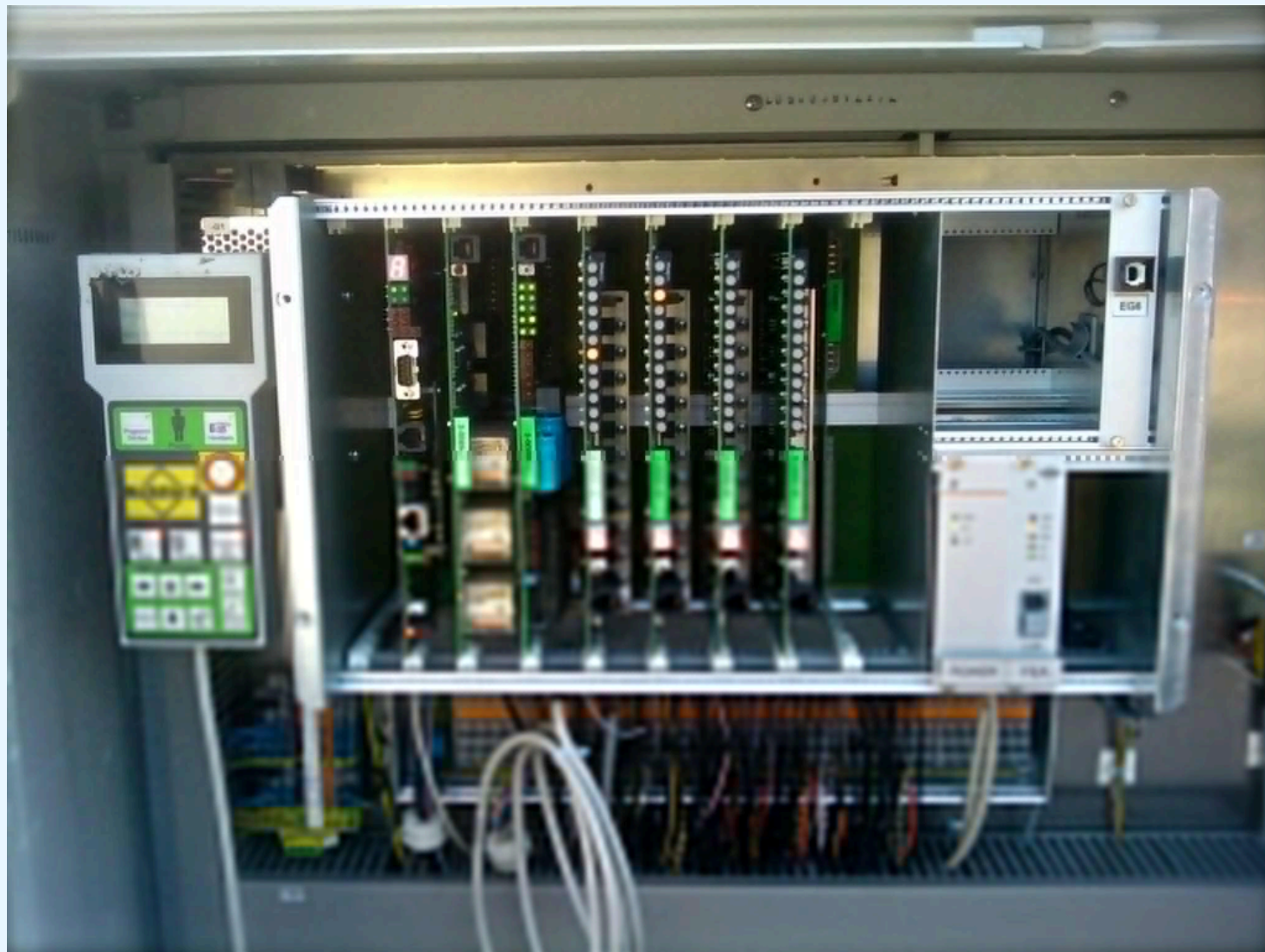
A Real-Time Operating System (RTOS) is an operating system that performs tasks within a specified time limit.

In traffic signal control, an RTOS helps manage traffic lights by controlling when red, yellow and green signals should turn ON and OFF.

Main Objective: To control traffic signals accurately, safely and on time.



TYPE OF OS USED AND JUSTIFICATION



Type of OS: Real-Time Operating System (RTOS)

An RTOS is designed to complete tasks within a fixed time limit.

Why is RTOS used in traffic signals?

- Time Management: Changes signals at the correct time.
- Fast Response: Responds quickly to sensor inputs.
- Task Scheduling: Manages multiple tasks.
- Reliability: Helps maintain proper signal operation.
- Safety: Prevents conflicting traffic signal instructions.

Example:

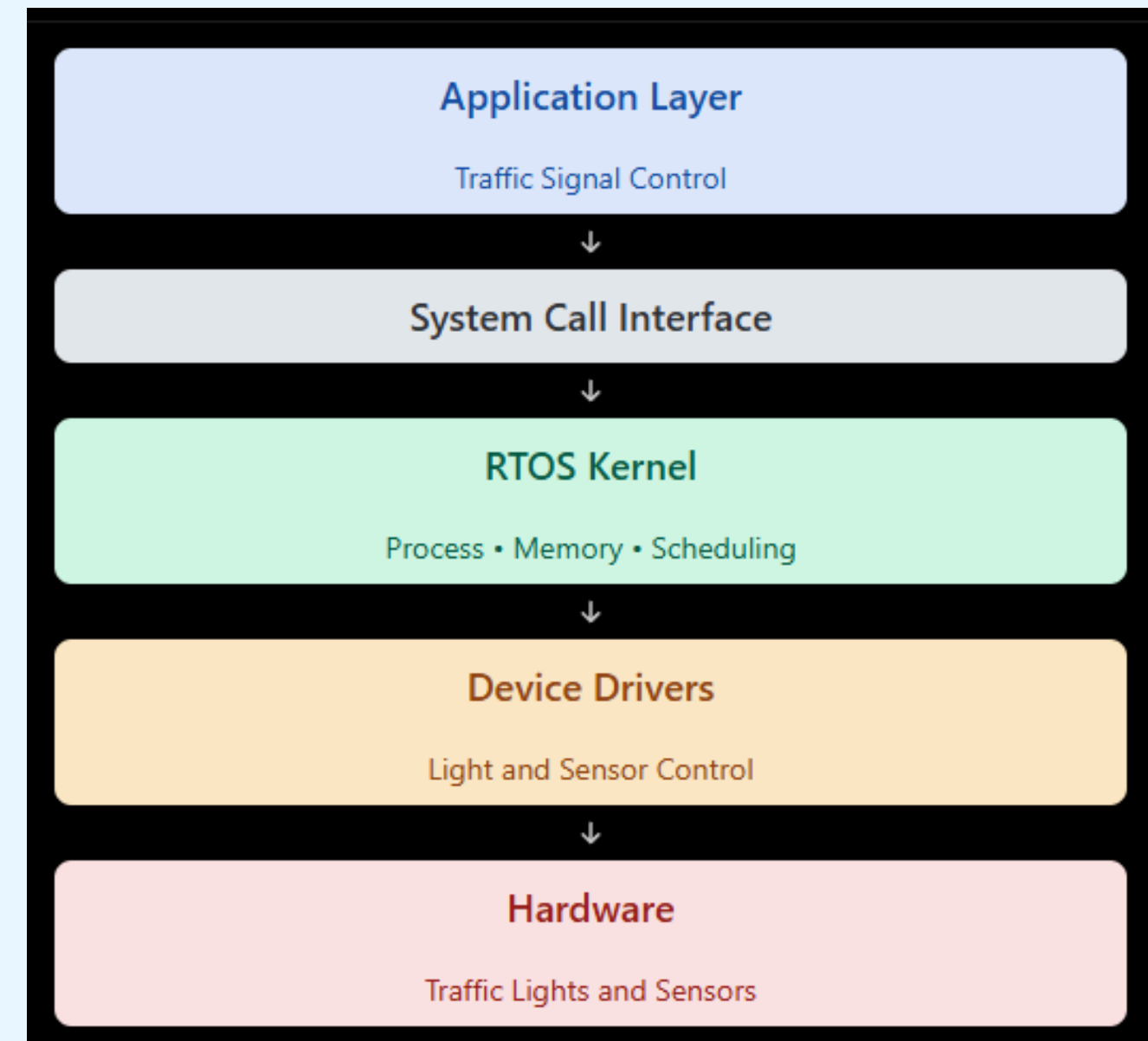
- Red light – Stop.
- Yellow light – Get ready.
- Green light – Go.

OS STRUCTURE AND COMPONENTS

OS Structure: Layered Architecture

The RTOS can be explained using the following components:

Component	Function
Application Layer	Controls traffic signal operations
System Call Interface	Connects applications with the kernel
Kernel	Manages all system activities
Process Management	Schedules traffic control tasks
Memory Management	Manages memory
Device Drivers	Control traffic lights and sensors
Hardware	Traffic lights, sensors and controller



SYSTEM CALLS USED FOR PROCESS CONTROL AND TIMING

What are System Calls?

System calls are functions that allow user programs to request services from the operating system kernel.

Important System Calls:

System Call	Purpose
<code>fork()</code>	Creates a new process (where supported)
<code>exec()</code>	Executes a program
<code>sleep()</code>	Pauses a process for a specified time
<code>wait()</code>	Waits for a process to complete
<code>clock_nanosleep()</code>	Provides timed sleeping on supported systems

Example in Traffic Signal Control:

Traffic Signal Timing Example		
RED 30 sec STOP	YELLOW 5 sec WAIT	GREEN 25 sec GO
Illustrative timing cycle (actual timings depend on traffic conditions).		



CONCLUSION & REFERENCES

- RTOS is suitable for traffic signal control.
- It performs tasks within specified time limits.
- It manages processes, memory and devices.
- System calls help control processes and timing.
- It improves traffic management and road safety.

References:

- Operating Systems Concepts – Abraham Silberschatz, Peter B. Galvin and Greg Gagne.
- IEEE Real-Time Systems concepts.
- POSIX System Calls Documentation.





THANK YOU!

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GPS Map Camera

Chennai, Tamil Nadu, India 

Thandalam Sriperumbadur, Chennai, Tamil Nadu
602105, India

Lat 13.026846, Long 80.017226

Thursday, 01/10/2026 01:58 PM GMT+05:30

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